

Comprehensive Final Exam

	Part I				(SO 1)		Sub total	Part II (Other)		
Q	1	2	3	4	5	6		7	8	9
Pts	/12	/12	/12	/12	/10	/12		/10	/10	/10
CLO	2	5	6	10	11	4		3	13	15

Answer all questions. Total points : 100

Part I – 70 points

1. Compile the following program to MIPS assembly. Use the variable-register mapping: x: \$t0 and base register of A: \$s0 and base register of B: \$s1. (CLO-2, SO-1) (12 pts)

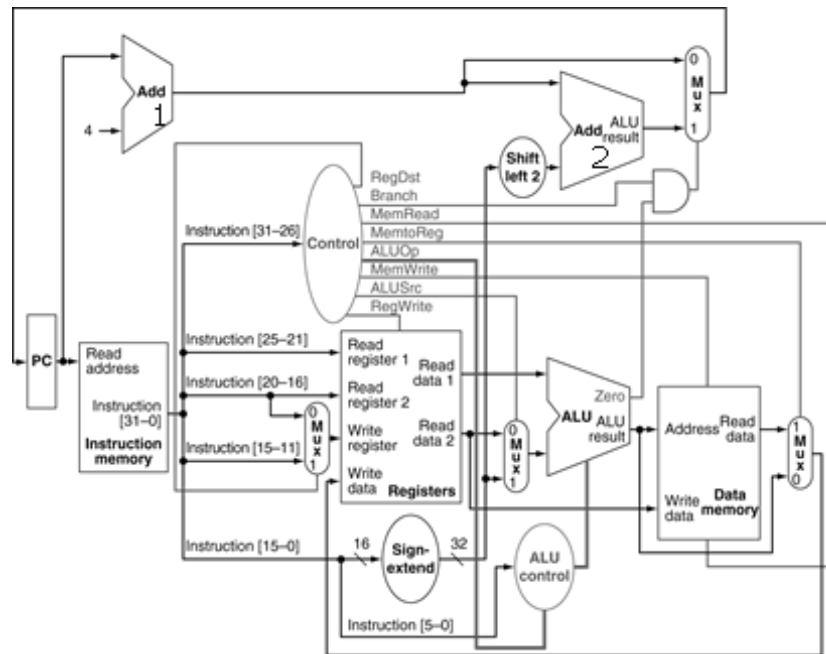
```
do{
    A[x+1] = 4*(B[x] + B[x+1]);
} while(int x=0; x<100; x++);
```

2. Consider that register \$s0 stores a number represented in the single-precision floating point number in the IEEE 754 format. Write MIPS code to identify its exponent (in decimal format) and store it in register \$t0. Set \$t0 to -999 if the number represents $\pm\infty$.

(CLO-5, SO-1) (12 pts)

3. The single-cycle 32-bit architecture is shown in the next figure.
SO-1) (12 pts)

(CLO-8,



Assume that the following instruction is executed on this architecture.

sw \$t0, 64 (\$s1)

- Is the constant 64 modified in anyway before it is sent to the ALU? If yes, what component of the diagram is used in this instruction? If no, what is modified? (4 points)
- Describe how ALU1, ALU2 and ALU (Main ALU) are used for this instruction? (4 points)
- What does the result of the ALU represent (as used in this instruction)? (2 points)
- What are the values of these control elements for this instruction? (2 points)

MemWrite	MemRead	RegWrite	ALUSrc

4. Describe the name, location and role of pipeline registers when executing instructions in a pipelined architecture. Describe the contents of the last pipeline register for the following instruction:

```
sw    $s2,    24 ($s3)
```

(CLO-10, SO-1) (12 pts)

5. In the following MIPS instruction sequence executing on a pipelined system: (CLO-11, SO-1) (10 points)

a. Which type of hazard is produced and at which instruction? (4 points)

```
Loop:  beq $t0, $t2, Exit
        add $s1, $s2, $s3
        sub $s2, $s1, $s3
        j  Loop
        ...
Exit:   ...
```

b. In general, what causes this type of hazard in a pipelined data path, and what are the ways that this can be eliminated? (6 points)

6. Write a procedure (method) that implements the given code in Java using MIPS code. Do not use pseudo-instructions. (CLO-4, SO-1) (12 pts)

```
int myFunc(int x, int y) {  
    if (y<=0)  
        return x%y;  
    else  
        return x*Math.pow(8, 5);  
}
```

Part II – 30 points

7. What are the contents of \$t0 and \$t2 after the four indicated instructions? (CLO-3, SO-1) (10 pts)

```

subi    $t0, $zero, -8           # instruction1
subi    $t2, $zero, 14          # instruction2
srl     $t2, $t2, 2              # instruction3
bne     $t2, $t0, ELSE
j       DONE

ELSE:   addi $t0, $t2, -3

DONE:                                     # instruction4

```

	\$t0	\$t2	Points
Instruction1		-	1 points
Instruction2			3 points
Instruction3			3 points
Instruction4			3 points

8. Describe how the index and tag are used in a 1024-block direct mapped cache. (CLO-13, SO-1) (10 pts)

9. Briefly explain the challenges of parallel programming. (CLO-15, SO-1) (10 points)